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4 **(a)** Fig. 4.1 shows a ray of light as it enters the side of a plastic block. The ray of light passes from air into the plastic.

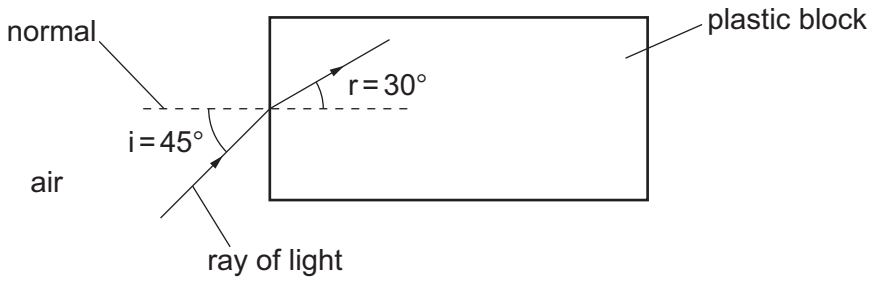

Fig. 4.1

(i) State how the speed, wavelength and frequency of the wave in the plastic block compare with their values in the air.

speed: .....

wavelength: .....

frequency: .....

[2]

(ii) Show that the refractive index of the plastic is 1.4.

Show your working.

[1]

(iii) Calculate the critical angle for the plastic.

critical angle = ... [2]

(b) Fig. 4.2 shows the same plastic as in **(a)** used to make an optical fibre. A ray of light is passing along the fibre.

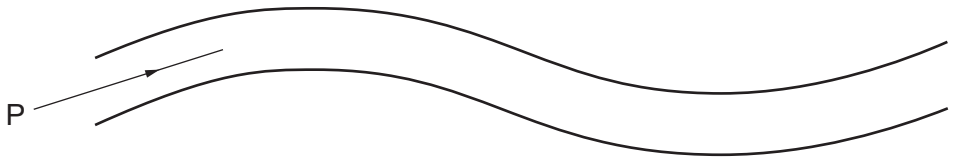

Fig. 4.2

(i) Carefully continue the ray of light P until it reaches the other end of the fibre. [2]

(ii) State **two** uses for optical fibres.

1 .....

2 .....

[2]

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